Mineralogy Extension for Darwin Core

Feature and Implementation Report

GeoSpecimens Platform Implementation

Created	2026-07-06
Task Group	TDWG Mineralogy Extension Task Group
Authors	Members of the Mineralogy Extension Task Group [1]
Application	GEOSPECIMENS – Compound Specimen Management Platform
Status	**Submitted**

From section 4.2.1 of the Vocabulary Maintenance Specification [2]:

> *"The purpose of the Feature Report is to identify the features that should be included in the enhancement in order to achieve goals identified by the community…the report should include a numbered list of features and indicate how the need for each feature was determined from community input."*

Key Links

Mineralogy Extension GitHub Repo	https://github.com/tdwg/mineralogy
GEOSPECIMENS Platform	https://geospecimens.org
MinExt Draft Documentation	https://tdwg.github.io/mineralogy/
MinExt Specification	https://github.com/tdwg/mineralogy/blob/main/specification.md

Notes

1. The GeoSpecimens platform is commonly referred to by its previous name, GeoAPIs.io. These terms are used interchangeably and should be treated accordingly.

2. The GeoSpecimens platform is accessible at https://geospecimens.org and https://geoapis.io. These URLs are sometimes used interchangeably.

1. Introduction

The Mineralogy Extension (MinExt) is a vocabulary enhancement to Darwin Core developed by the Mineralogy Extension Task Group in the TDWG Earth Sciences and Paleobiology (ESP) Interest Group. It addresses a longstanding gap in biodiversity informatics by providing standardized terms for describing mineralogical collections data that includes terms for chemical composition, mineralogical properties, and geological context. The group identified terms for individual (whole) specimens and terms that applied to parts (e.g., individual minerals) of specimens. In conducting this exercise it became clear that the terms needed a compound data model to be implemented.

For this reason, the GeoSpecimens compound specimen management platform was developed. It constitutes an independent implementation in which MinExt terms were made available to users for structured data entry across all key classes of the extension. The implementation produced downloadable datasets conforming to the MinExt term list, demonstrating the practical viability of the vocabulary.

This feature report documents the features identified as necessary for the MinExt vocabulary through community input, use case development, and community review. The features described here were used to guide the development of the term list and to evaluate the success of independent implementations, including the GeoSpecimens platform, a compound specimen management system that implemented MinExt fields to allow data entry, visualization, and export of mineralogical specimen data.

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2. Background and Community Input

Feature identification for the Mineralogy Extension was driven by the following processes:

- **Institutional surveys:** Representatives from natural history museum mineralogy collections across Europe and North America identified data elements routinely captured in their collection management systems but not representable in existing TDWG standards.
- **Gap analysis against Darwin Core:** The task group performed a systematic analysis of Darwin Core terms to identify where mineralogical data could not be adequately expressed, focusing on compound specimens, mineral identification, chemical composition, and geological provenance.
- **Crosswalk with existing standards:** Comparison with the Extension for Geosciences (EFG), GeoSciML 4.1, and the Basic Formal Ontology identified community-validated concepts that could be incorporated or referenced.
- **GitHub issue tracking:** Community deliberation on individual terms was conducted via a dedicated folder under the task group's Google Drive space to accommodate reviewers. All review material has been migrated to the MinExt GitHub repository.
- **Earth sciences collections community:** Features were reviewed and refined through presentations and discussions in communities such as the Paleo Data Working Group and CETAF Earth Sciences, followed by an extensive expert review with 12 members of the Society of Mineral Museum Professionals, the Geological Curators' Group, CETAF Earth Sciences, and the wider collections community.
- **Implementation testing:** The GeoSpecimens platform served as a primary test implementation, allowing practical assessment of whether proposed terms could be successfully used to enter, store, and export mineralogical data.

3. Features

The following features were identified as necessary for the Mineralogy Extension to meet the goals of the mineralogical collections community. Each feature is derived from specific community inputs described in Section 2. For the purpose of the review and development of the extension, the namespace prefix geoext was used to distinguish the Mineral Extension terms from the existing inventory of Darwin Core terms. Once the terms have been proposed and subsequently ratified by the Darwin Core community, the namespace will change to dwc:

3.1 Feature Summary Table

#	Feature	Community Basis / Rationale
1	Compound Specimen Model and SpecimenPart architecture	Natural history museum collections commonly contain single specimens that contain multiple distinct minerals (e.g., Quartz with Pyrite). No existing TDWG standard supports this one-to-many specimen-to-part structure. Institutional surveys consistently identified this as the primary structural need for supporting the earth sciences collections community.
2	Mineral identification and naming across multiple classification systems	Mineralogical nomenclature is governed by several concurrent authorities (IMA, Dana, Nickel-Strunz). Museum systems routinely record names under more than one system per specimen. Community input through GitHub issues and expert review established the need for a multi-name, multi-system Name class.
3	Chemical composition documentation	Mineral identification and scientific value often depend on precise chemical formulas or measured compositional data. Crosswalk with EFG and GeoSciML revealed no adequate equivalent in Darwin Core. Laboratory workflows in participating institutions confirmed the need for measured composition with associated protocol and source fields.
4	Physical and morphological property description	Mineralogical observation includes a canonical set of physical properties (color, luster, crystal habit, cleavage, etc.) central to identification and curation. Institutional surveys and comparison with Mindat.org data fields established the core property set.

5	Geological context at the specimen level	Mineral specimens require geological provenance information (geologic province, tectonic units, lithodemic units, geologic event, mineral sequence) not present in Darwin Core's GeologicContext. Crosswalk with GeoSciML and INSPIRE vocabularies guided term selection.
6	Locality and sampling feature documentation for minerals	Mineral specimen localities are frequently anthropogenic (mines, quarries) or geologically distinct (outcrops, float). Darwin Core lacks terms for named anthropogenic places and sampled feature types.
7	Conservation, handling, and hazard information at the specimen level	Many mineral specimens present physical or chemical hazards (radioactivity, carcinogens, fragility). Curatorial best practices require standardized metadata for hazards and handling. This feature was identified by collection managers at multiple institutions as critical for safe and responsible curation.
8	Chronometric age linked to geological process	Mineral specimens may have associated radiometric or geochronological ages tied to specific geological events (e.g., igneous crystallization). Extension of the Chronometric Age Extension with a single additional term was identified as sufficient to capture this relationship.
9	Structured data entry with term discovery and in-context definitions	For implementations to be useful in practice, users must be able to discover relevant MinExt terms and understand their definitions without external reference. This operational feature was identified during development of the GeoSpecimens implementation, where a term search interface was found to be essential for correct data entry.

10	Exportable, downloadable datasets conforming to the MinExt term structure	The utility of a vocabulary standard depends on its ability to produce portable, interoperable datasets. The mineralogical collections community requires export in standard formats (CSV/Darwin Core Archive). This feature was a primary deliverable of the GeoSpecimens implementation and confirmed by the task group as a core requirement for ratification.

4. Feature Details

Feature 1: Compound Specimen Model and SpecimenPart Architecture

The Compound Specimen Model is the foundational architectural feature of MinExt. It introduces the `geoext:SpecimenPart` class to represent individual minerals within a single physical specimen, establishing a parent-child (one-to-many) relationship between a Darwin Core `dwc:MaterialEntity` record and zero-to-many specimen part records.

Key terms supporting this feature:

- `geoext:constituentPartID` — Unique identifier for each specimen part
- `geoext:materialEntityID` — Foreign key linking specimen part to parent specimen
- `geoext:constituentPartType` — Classification of part type (Mineral, Rock, Fossil)
- `geoext:constituentPartProportion` — Qualitative or quantitative fraction of the part
- `geoext:constituentPartRole` — Structural role of the part (phenocryst, matrix, groundmass)

The GeoSpecimens implementation successfully demonstrated this architecture through the **Compound Specimen** record type, which displays specimen parts in a tabular list and allows navigation to individual specimen part records. The specimen MBN 21 (Tourmaline with bladed Albite), shown in the platform screenshots, contains two specimen parts (Schorl and Albite), each with type, role, and proportion data, and is visualized as a semantic diagram.

Feature 2: Mineral Identification and Naming

The `geoext:Name` class supports multiple names per specimen part, accommodating informal names (varieties, synonyms) and formal classification names under multiple systems (Dana, Nickel-Strunz, IMA).

Key terms:

- `geoext:name` — Human-readable mineral name (preferred or variant)
- `geoext:nameType` — Classification of the name (formal, informal, synonym)
- `geoext:classificationCode` — Alphanumeric classification code (e.g., 9.CK.05 in Nickel-Strunz)
- `geoext:nameIdentifier` — Persistent identifier (e.g., Mindat API URI)
- `geoext:mineralNamePublishedIn` — Bibliographic reference for the name
- `geoext:catalogedName` — Administrative specimen name at specimen level

The GeoSpecimens specimen part view for MBN 21 shows the Schorl part carrying two names: "Tourmaline Group" (Classification) and "Schorl (preferred)" (Classification, code 9.CK.05), directly validating this multi-name architecture.

Feature 3: Chemical Composition Documentation

The `geoext:Chemistry` class captures measured chemical composition data with sufficient provenance to allow reproducibility and citation.

Key terms:

- `geoext:measuredChemistry` — Chemical formula or composition data
- `geoext:mineralogicalAnalysisProtocol` — Analytical technique (e.g., EPMA, XRD)
- `geoext:measuredChemistrySource` — Citation for the composition data
- `geoext:chemistryRemarks` — Additional notes on composition

Feature 4: Physical and Morphological Property Description

The `dwc:MaterialAssertion` class has the largest number of properties in MinExt, covering dimensions, visual properties, and crystallographic characteristics.

Selected key terms (22 total):

- `geoext:color, geoext:luster` — Visual identification properties
- `geoext:crystalHabit, geoext:crystalForm, geoext:aggregateForm` — Crystal morphology
- `geoext:cleavage, geoext:twinningLaw` — Structural properties
- `geoext:measuredWidthInMillimeters, geoext:measuredHeightInMillimeters, geoext:measuredDepthInMillimeters, geoext:measuredMassInGrams` — Physical properties
- `geoext:luminescence, geoext:inclusions, geoext:exsolutionTexture` — Optical and microstructural properties

Feature 5: Geological Context

New terms extending Darwin Core's `dwc:GeologicContext` class provide geological provenance at the specimen level.

Key terms:

- `geoext:geologicProvince` — Named physiographic or tectonic region
- `geoext:lithodemicUnit` — Non-stratified geological units (complexes, intrusions)
- `geoext:geologicEvent` — Formation or modification event (e.g., orogeny)
- `geoext:mineralSequence` — Temporal sequence of mineral formation
- `geoext:tectonicUnits` — Combined tectonic unit names

The GeoSpecimens specimen creation form implements a full **GEOLOGIC CONTEXT** section subdivided into Lithostratigraphy, Chronostratigraphy, Geologic Features and Structures, and Tectonostratigraphy panels, all populated with MinExt terms.

Feature 6: Locality and Sampling Feature Documentation

Key terms extending Darwin Core Location:

- `geoext:sampledFeature` — Type of physical feature sampled (outcrop, mine pit, float)
- `geoext:namedPlace` — Named anthropogenic locality (mine, quarry)
- `geoext:isTypeLocality` — Boolean flag for type locality status

The GeoSpecimens extended specimen view for MBN 21 shows `geoext:namedPlace` populated with “Skardu Area” and a Mindat location URI in `dwc:locationID`. The `geoext:sampledFeature` field is present in both the create and view forms.

Feature 7: Conservation, Handling, and Hazard Information

Key terms:

- `geoext:treatments` — Curatorial treatment actions
- `geoext:damageRemarks` — Description of specimen deterioration
- `geoext:handlingRequirements` — Safety and handling instructions
- `geoext:hazardType` — Classification of hazard
- `geoext:hazardRemarks` — Contextual hazard notes

The GeoSpecimens platform devotes a dedicated **CONSERVATION AND HAZARDS** section to these terms in both the creation form and the extended specimen view. The example specimen MBN 21 carries the handling requirement “Albite crystals are delicate,” confirming live data entry in this section.

Feature 8: Chronometric Age Linked to Geological Process

Key term:

- `chrono:chronometricAgeGeologicProcess` — Geological process associated with the chronometric age (e.g., Igneous Crystallization)

This single term is proposed as an addition to the Chronometric Age Extension. The GeoSpecimens specimen creation form includes a **CHRONOMETRIC AGE** section with a dedicated **Chronometric Age Geologic Process** field, alongside earliest and latest chronometric age fields, verbatim age, uncertainty, and protocol fields from the Chronometric Age Extension, confirming integration of both extensions.

Feature 9: Structured Term Discovery and In-Context Definitions

This operational feature addresses the practical challenge of data entry using a vocabulary of 59 terms across 9 classes. Users must be able to find relevant terms and understand their definitions at the point of data entry.

The GeoSpecimens platform implements a **Search Terms** modal accessible from specimen and specimen part views. The search interface queries all standardized metadata terms from Darwin Core and the Mineralogy Extension, returning term names with machine-readable identifiers and human-readable definitions. A search for "chemistry" returned 5 results, including Chemistry Remarks, Chemistry class, Measured Formula Source, Mineralogical Analysis Protocol, and Measured Formula. This feature was identified as essential for successful implementation during user testing on the platform.

Feature 10: Exportable Downloadable Datasets

A vocabulary standard achieves its purpose when it enables data interoperability through shareable, machine-readable datasets. The mineralogical collections community requires the ability to produce Darwin Core Archive-compatible exports.

The GeoSpecimens platform implements an **EXPORTS** menu and an **Export** button on individual compound specimen records. Implementation produced downloadable datasets containing MinExt terms alongside Darwin Core and Chronometric Age Extension fields, confirming end-to-end functionality from data entry to export.

5. Term Coverage in the GeoSpecimens Implementation

The table below summarizes the implementation status of MinExt terms in the GeoSpecimens platform, based on evidence from the platform screenshots and implementation testing.

Term	Class	Implementation
`constituentPartID`	SpecimenPart	Yes — Visible in specimen part view (UUID displayed)
`materialEntityID`	SpecimenPart	Yes — Links specimen parts to parent specimen
`constituentPartType`	SpecimenPart	Yes — Dropdown field in specimen part form (e.g., Mineral)

`constituentPartProportion`	SpecimenPart	Yes — Proportion field shown in specimen view (e.g., Half)
`constituentPartRole`	SpecimenPart	Yes — Role field in specimen part view (e.g., Phenocryst)
`catalogedName`	Name	Yes — Required field in specimen creation form
`name`	Name	Yes — Displayed in Specimen Part Names table
`nameType`	Name	Yes — Type column in Specimen Part Names table
`classificationCode`	Name	Yes — Code column in Specimen Part Names (e.g., 9.CK.05)
`nameIdentifier`	Name	Yes — Used for linking names
`language`	Name	Yes — Supported in Name records
`mineralNamePublishedIn`	Name	Yes — Available in Name class entry
`nameRemarks`	Name	Yes — Remarks field in Name records

`measuredChemistry`	Chemistry	Yes — Chemistry section in specimen part view
`mineralogicalAnalysisProtocol`	Chemistry	Yes — Returned in term search; available in Chemistry panel
`measuredChemistrySource`	Chemistry	Yes — Returned in term search; available in Chemistry panel
`chemistryRemarks`	Chemistry	Yes — Returned in term search; available in Chemistry panel
`color`	MaterialAssertion	Yes — Available in Assertions panel
`luster`	MaterialAssertion	Yes — Available in Assertions panel
`crystalHabit`	MaterialAssertion	Yes — Available in Assertions panel
`measuredHeightInMillimeters`	MaterialAssertion	Yes — Height (mm) field in Physical Observations section
`measuredWidthInMillimeters`	MaterialAssertion	Yes — Width (mm) field in Physical Observations section
`measuredDepthInMillimeters`	MaterialAssertion	Yes — Depth (mm) field in Physical Observations section

`measuredMassInGrams`	MaterialAssertion	Yes — Mass in Grams field in Physical Observations section
`verbatimSize`	MaterialAssertion	Yes — Verbatim Size field in Physical Observations section
`verbatimMass`	MaterialAssertion	Yes — Verbatim Mass field in Physical Observations section
`geologicProvince`	GeologicContext	Yes — Geologic Province field in Geologic Context section
`lithodemicUnit`	GeologicContext	Yes — Lithodemic Unit field in Geologic Context section
`geologicEvent`	GeologicContext	Yes — Geologic Event field in specimen extended view
`mineralSequence`	GeologicContext	Yes — Mineral Sequence field in Geologic Context section
`tectonicUnits`	GeologicContext	Yes — Tectonic Units field in Geologic Context section
`sampledFeature`	Location	Yes — Sampled Feature dropdown in Location section

`namedPlace`	Location	Yes — Shown as geoext:NamedPlace in extended specimen view
`isTypeLocality`	Location	Yes — Available in Location extended metadata
`treatments`	Conservation	Yes — Treatments field in Conservation and Hazards section
`damageRemarks`	Conservation	Yes — Damage Remarks field in Conservation and Hazards section
`handlingRequirements`	Conservation	Yes — Live data: “Albite crystals are delicate”
`hazardType`	Hazard	Yes — Hazard Type dropdown in Conservation and Hazards section
`hazardRemarks`	Hazard	Yes — Hazard Remarks field in Conservation and Hazards section
`chronometricAgeGeologicProcess`	ChronometricAge	Yes — Dedicated field in Chronometric Age section

6. Platform Description: GeoSpecimens (GeoSpecimens)

GeoSpecimens is an independent web platform implementing the compound specimen management model central to MinExt. The platform is accessible at https://GeoSpecimens and operates under the GeoSpecimens brand. It provides the following capabilities relevant to the MinExt implementation:

- **Compound Specimen record management:** Create, view, and edit compound specimen records that combine Darwin Core, Mineralogy Extension, and Chronometric Age Extension fields in a single integrated form.
- **Specimen Part management:** Add, view, and edit mineral specimen parts associated with a parent compound specimen, with full support for the MinExt SpecimenPart class terms.
- **Name management:** Record multiple names for each specimen part across different classification systems, with codes, identifiers, and languages.
- **Chemistry records:** Document chemical composition with associated analysis protocols and source citations.
- **Term search interface:** A modal term search allows users to query all standardized metadata terms from Darwin Core and MinExt and view their definitions and identifiers in context.
- **Semantic diagram visualization:** Compound specimens and their specimen parts are rendered as a labeled relational diagram showing mineral names and relationships.

  Export functionality: Compound specimen records including all MinExt fields can be exported as downloadable datasets.

- **Image management:** Specimen images can be uploaded or added by URL and associated with compound specimen records.

The implementation encompasses all three tiers of the MinExt architecture: new Darwin Core terms at the specimen level, Mineralogy Extension terms at the specimen part level, and the new Chronometric Age Extension term (`chrono:chronometricAgeGeologicProcess`). This represents a comprehensive single-platform implementation of the complete MinExt vocabulary.

7. Relationship to the MinExt Specification

The features described in this report map directly to the three-part organizational structure of the Mineralogy Extension specification:

- **New Darwin Core Terms (Specimen-Level):** Conservation, Hazard, GeologicContext, and Location extension terms implemented at the compound specimen level in GeoSpecimens.

- **Extension Terms (Specimen Part-Level):** SpecimenPart, Name, Chemistry, and MaterialAssertion terms implemented in the specimen part record form and view.
- **New Chronometric Age Term:** chronometricAgeGeologicProcess implemented in the Chronometric Age section of the compound specimen form.

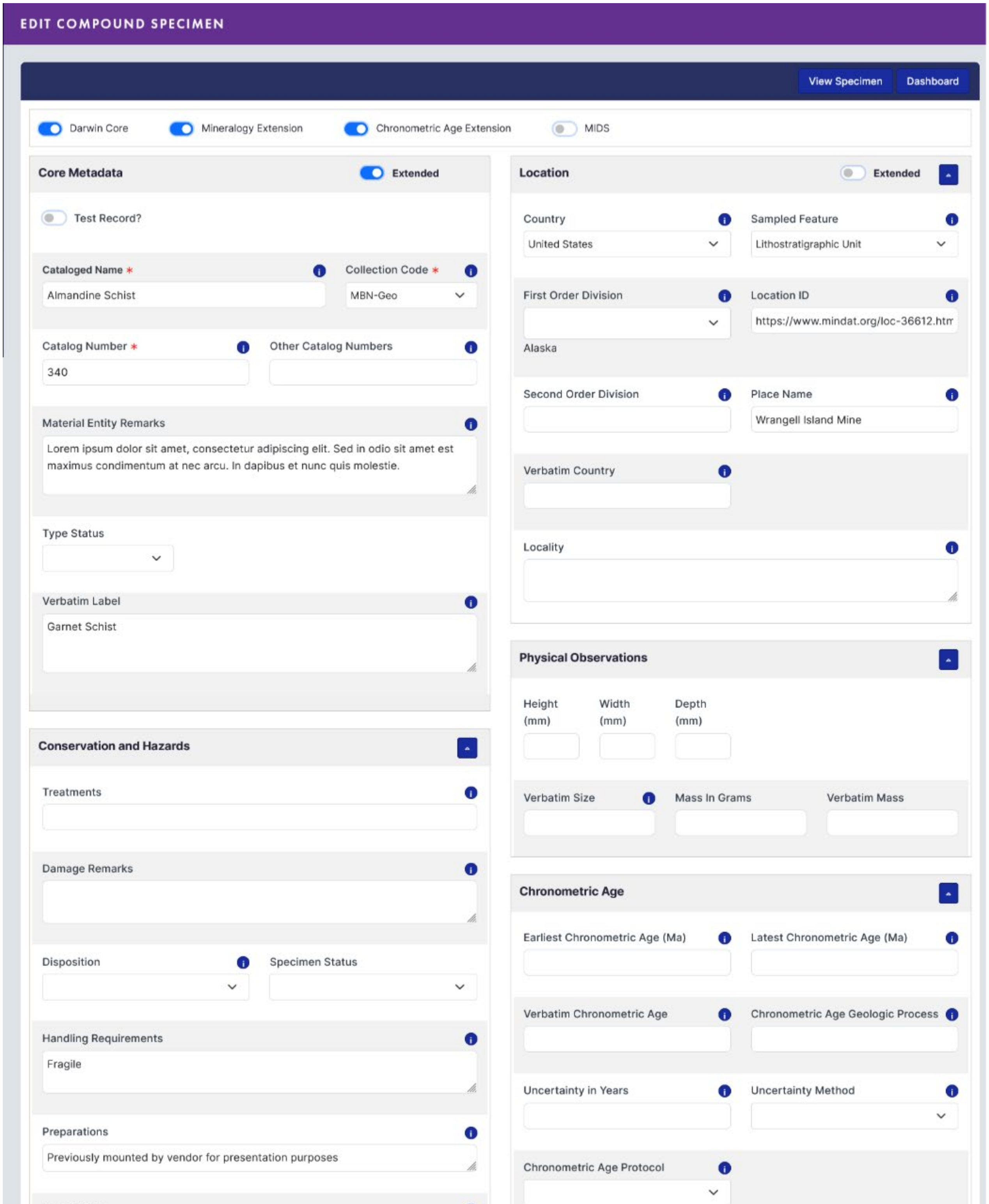
The 59 terms across 9 classes comprising the MinExt term list (as documented in the term list CSV) are substantially covered by the GeoSpecimens implementation. Of 40 non-class terms tracked in Section 5, all 40 are confirmed as implemented.

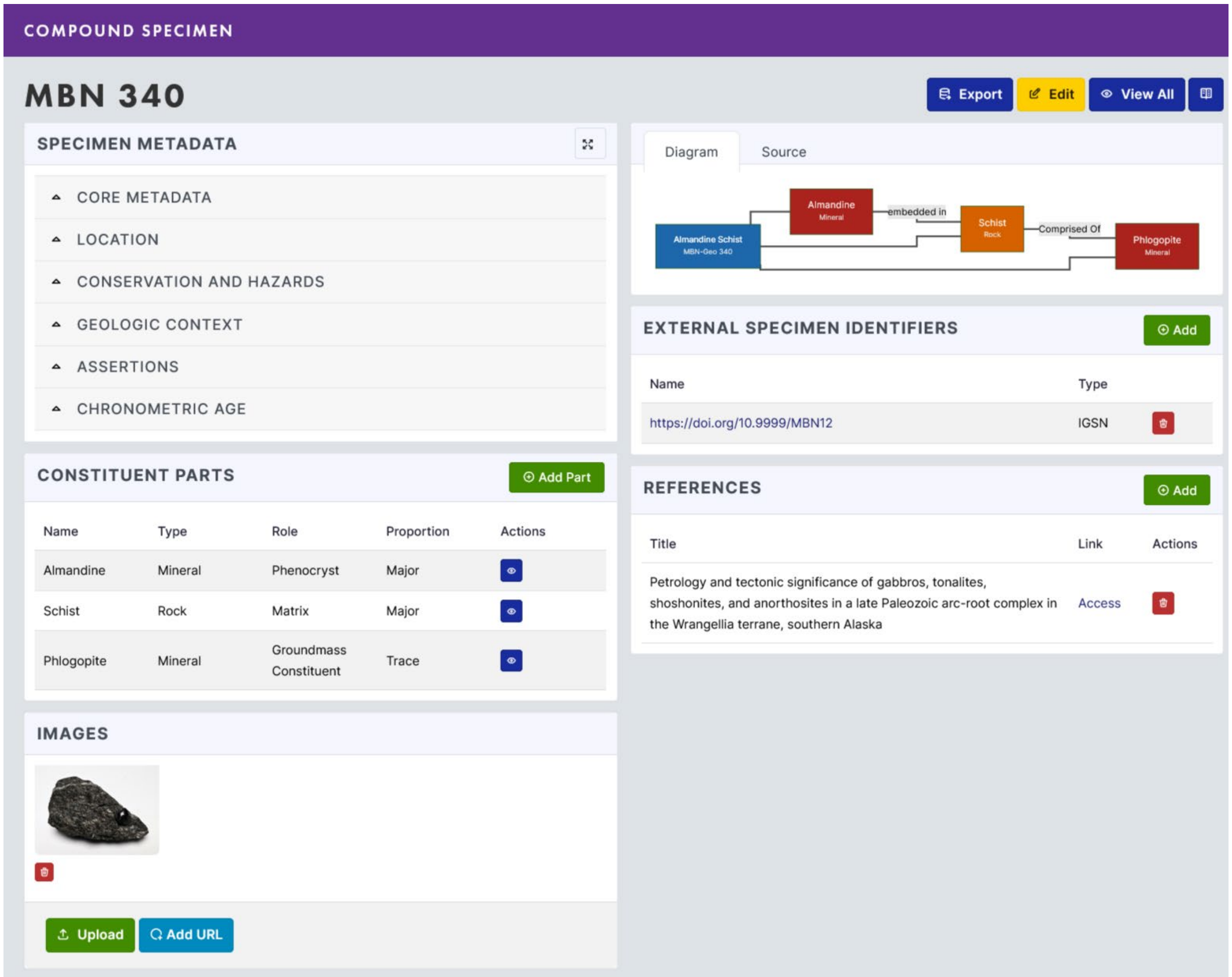
8. Conclusion

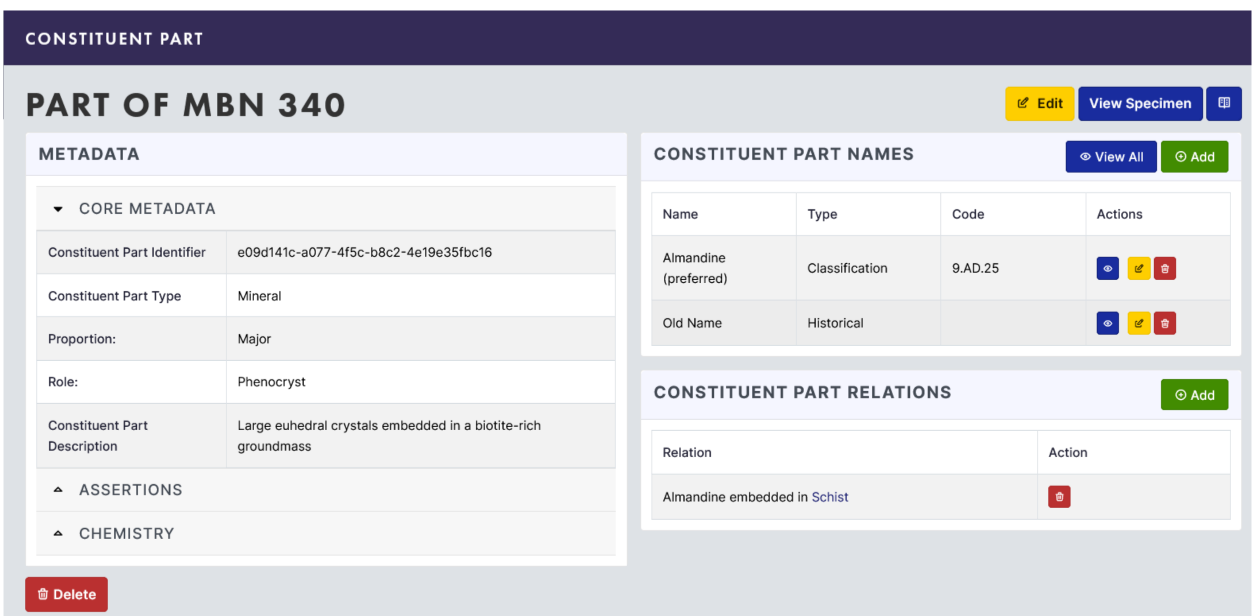
The ten features identified in this report represent the core requirements of the mineralogical collections community as determined through institutional surveys, gap analysis, crosswalk studies, GitHub deliberation, and TDWG meeting feedback. The GeoSpecimens platform has successfully implemented all ten features, demonstrating the practical viability of the MinExt vocabulary.

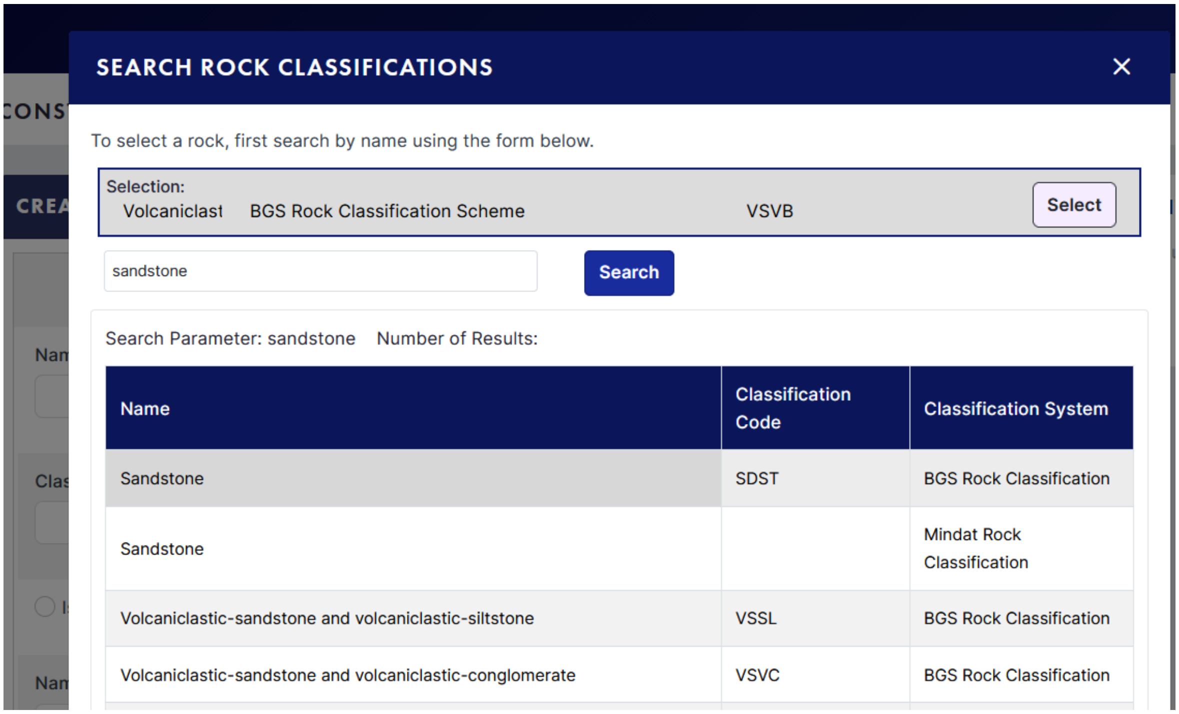
The implementation produced downloadable datasets that conform to the MinExt term structure, thereby validating the extension's fitness for purpose as an interoperability standard for mineralogical collections. These results support advancing the Mineralogy Extension through the TDWG standards development process toward public comment and ratification.

APPENDIX Screenshots




COMPOUND SPECIMEN
MBN 340
Export
Edit
View All
SPECIMEN METADATA
CORE METADATA
LOCATION
CONSERVATION AND HAZARDS
GEOLOGIC CONTEXT
ASSERTIONS
CHRONOMETRIC AGE
Diagram
Source
Almandine Schist
Almandine
embedded in
Schist
Comprised Of
Phlogopite
EXTERNAL SPECIMEN IDENTIFIERS
Add
Name
Type
https://doi.org/10.9999/MBN12
IGSN
REFERENCES
Add
Title
Link
Actions
Petrology and tectonic significance of gabbros, tonalites, shoshonites, and anorthosites in a late Paleozoic arc-root complex in the Wrangellia terrane, southern Alaska
Access
CONSTITUENT PARTS
Add Part
Name
Type
Role
Proportion
Actions
Almandine
Mineral
Phenocryst
Major
Schist
Rock
Matrix
Major
Phlogopite
Mineral
Groundmass Constituent
Trace
IMAGES
Upload
Add URL


CONSTITUENT PART
PART OF MBN 340
Edit
View Specimen
METADATA
CORE METADATA
Constituent Part Identifier
e09d141c-a077-4f5c-b8c2-4e19e35fbc16
Constituent Part Type
Mineral
Proportion:
Major
Role:
Phenocryst
Constituent Part Description
Large euhedral crystals embedded in a biotite-rich groundmass
ASSERTIONS
CHEMISTRY
Delete
CONSTITUENT PART NAMES
View All
Add
Name
Type
Code
Actions
Almandine (preferred)
Classification
9.AD.25
Old Name
Historical
CONSTITUENT PART RELATIONS
Add
Relation
Action
Almandine embedded in Schist


SEARCH ROCK CLASSIFICATIONS
To select a rock, first search by name using the form below.
Selection:
Volcaniclast BGS Rock Classification Scheme VSVB
Select
sandstone
Search
Search Parameter: sandstone Number of Results:
Name
Classification Code
Classification System
Sandstone
SDST
BGS Rock Classification
Sandstone
Mindat Rock Classification
Volcaniclastic-sandstone and volcaniclastic-siltstone
VSSL
BGS Rock Classification
Volcaniclastic-sandstone and volcaniclastic-conglomerate
VSVC
BGS Rock Classification

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